Part-2: Supply Requirements

Section 6 – Schedule of Supply

The Schedule of Supply consists of five parts:

1-1. List of Goods and Related Services
1-2. Delivery and Completion Schedule

2. Technical Specifications
3. Drawings
4. Inspections and Tests
5. PACKING, MARKING AND DRAWINGS INSTRUCTIONS FOR BOX DESIGN

Detail Address of Destination as below:

Sl No.	Name of Warehouse	Address
01	Central Warehouse, Khulna.	Shiromoni, Khulna, Bangladesh.

1-1. List of Goods and Related Services

Name of Bidder _______________
Item: 33 kv/11 kv ACR , Unit in Nos.
Package Number – MCEP/BREB/KD-G-20, No. MCEP/BREB/KD-G-20-01, Page ___ of ___

Line Item No.	Description of Goods	Quantity	Physical unit	Delivery (to the final destination) period from the date of contract signing				
				Quantity to deliver at Final Destination (Project Sites)	Latest Delivery Date (1st phase)*	Latest Delivery Date (2nd phase)*	Bidder's offered Delivery date [to be provided by the bidder]	
							1st phase	2nd phase
1	2	3	4	5	6	7	8	9
HS-7.800B	33KV,800 Amps, Automatic Circuit recloser 3 phase 50hz	6	Nos.	Destination Places are as per Table 1A-i (Details Delivery Schedule)	100% of each item within 20 weeks from the date of contract signing	NA	[Insert the number of weeks following the date of Contract signing]	NA
HS-8.400B	11KV, 400 Amps, Automatic Circuit recloser, 3 phase 50hz	28	Nos.					

Name of Bidder:
Signature of Bidder:
Date:

1-2 Delivery and Completion Schedule

1A-i Details Delivery Schedule for : 33 kv/11 kv ACR

Name of Bidder _______________
Item: 33 kv/11 kv ACR, Unit in Nos.
Package Number – MCEP/BREB/KD-G-20, No. MCEP/BREB/KD-G-20-01, Page ___ of ___

100% of the quantity against each item at BREB's Central Warehouse ,Shiromoni, Khulna, Bangladesh

2. Technical Specifications

PUBLICATION 308B-2010
RURAL ELECTRIFICATION BOARD (REB)
PEOPLES REPUBLIC OF BANGLADESH
STANDARD FOR CLASS III
33 KV THREE PHASE SOLID DI-ELECTRIC
AUTOMATIC CIRCUIT RECLOSERS

1. **SCOPE:**

This standard covers the electrical, mechanical and performance requirements for three phase, automatic circuit reclosers for use on the REB overhead distribution system and in REB 33/11 KV substations.

2. **GENERAL:**

Reclosers furnished to REB specifications shall conform in all respects to the requirements of this standard. These reclosers shall be suitable for use in a tropical climate with high relative humidity up to 95%. The text, tables, figures and performances to other standards supplement each other and are considered part of this standard.

3. **RATINGS:**

3.1 Service Conditions:

a. The required operating temperature range shall be between-10C to + 65^0 C.
b. Altitude-up to 1,000 meters.

3.2 Rated Voltage

a. Nominal system voltage-33,000 volt source (Delta)
b. Rated maximum voltage - 38,000 volts.
c. Low frequency withstand voltage, RMS

i) Dry, 60 seconds-70 KV
ii) Wet,10 seconds-60 KV

3.3 Rated frequency - 50 Hertz.

3.4 Rated minimum current shall be as shown in Table I.

3.5 Rated minimum tripping current shall be as shown in Table-I with a tolerance of ± 5%.

3.6 Rated symmetrical interrupting current shall be as in Table-I with asymmetrical relationships as defined in ANSI/IEEE C37.60.

3.7 Rated symmetrical making current shall be the same value as the rated symmetrical interrupting current, with asymmetrical relationships as defined in ANSI/IEEE C37.60.

3.8 Rated minimum Impulse Withstand Voltage - 150 KV (min).

4. **DESIGN**

Design of the REB class III recloser shall be in accordance with ANSI/IEEE C37.60. The recloser shall be designed to provide self-contained distribution circuit protection with series trip, magnetic actuators and electronic control double or single break contact structure and fault interruption under vacuum. The vacuum interrupters shall be in accordance with ANSI/IEEE C37.85. The recloser shall utilise vacuum interrupter as the interruption medium and solid dielectric as the insulation medium. The recloser shall be designed to sense line-to-ground, line-to-line or three-phase fault currents and interrupt all three phases of the distribution circuit simultaneously. The recloser shall be equipped with an electronic type ground trip. If the Electronic control cubicle of Reclosers or Recloser itself require an auxiliary transformer, the manufacturer must provide the same at his cost. The recloser shall be provided with in-built auxiliary transformer of required rating to provide power supply to the control panel. The secondary voltage of the said PT must be 240V.

For 33KV solid dielectric ACR, terminal to terminal and lower terminal to ground/earth clearance shall not be less than 394 mm and 676 mm respectively. Terminal to terminal and lower terminal to ground/earth creepage distance shall not be less than 1763 mm and 1215 mm respectively.

Design features are summarised below:

4.1 The Recloser head shall be fabricated of corrosion resistant materials or provided with an impact and corrosion resistant finish and should also be suitable for storage in uncovered areas and shall be provided with the following features:

a: Universal clamp type terminals which will accept #6 AWG to 350 MCM copper or aluminium conductors either horizontally or vertically. In place of universal clamp, insulated and water blocked cable tails of proper voltage and current rating is acceptable.

b: Silicon bushings with bushing clamps bolted to head with bird (guard) protection or epoxy bushing and EPDM rubber bushing boots to provgide a fully insulated bushing arrangement.

c: Lifting provision is required to lift the entire recloser so positioned that the recloser will remain level when lifted.

d: Manual Operating Provision: Recloser shall be provided with a manual operating lever, which is suitable for operating with a hot stick or other suitable manual operating means.

e: Non-reclosing lever/feature to set recloser to lock out after one operation.

f: Counters: An operation counter shall be provided to indicate the total number of operations of the reclosers as well as the number of fault operations for each type of fault . The counter shall be visible with recloser in service.

g: Attachment clamps and an "0-ring" seal to fix tank to head in such a manner as to provide a positive, moisture-proof seal.

h: Closing tool port for manually closing de-energized recloser, Provisions should be made for manually closing de-energized recloser without un-tanking it. A manual closing tool shall be provided with each recloser. Alternatively closing and tripping by LV solenoid and control cubicle is acceptable.

i: Nameplate(s) listing factory ratings and settings of the recloser, with provisions for changing of nameplate data when field changes are made affecting operating characteristics of the recloser. Nameplate information shall include, but not be limited to the following:

Manufacturer's Identification.

Contract number and IFB number

Recloser type.

Recloser serial number.

Rated maximum voltage.

Rated minimum continuous current.

Rated minimum tripping current.

Rated minimum symmetrical interrupting current.

Rated minimum impulse withstand voltage.

Number of fast operations & also identification of Operational Curve.

Number of slow operations & also identification of Operational Curve.

Total operations to lock-out.

Designation of operating curves.

BIL Rating.

Nominal system voltage.

Year of manufacture.

4.2 The recloser tank shall be fabricated of corrosion resistant steel, and provided with an impact and corrosion resistant finish and should also be suitable for storage in uncovered area and provided with a mating surface for the head so that leaks will not occur and that the recloser will remain mechanically operable at the maximum operating pressure generated by normal operation of the recloser. The tank shall have the following features:

a. Mounting lugs for mounting the recloser in a substation frame or cross arm mounting hanger.

b. Ground connector to accept #8 to #2/0 AWG copper or aluminum conductors.

c. The finish shall be a thermosetting impact and corrosion resisting, acrylic paint (ANSI Colour #70) over inhibiting epoxy prime coat.

d. A 10(ten) meter long control cable should be provided with ACR for mounting the operator control panel with meters at lower position of the pole. Pole mounting clamps/hanger for control panel should be provided.

4.3 The electronic ground trip device shall be self-contained and independent. REB three phase recloser shall be equipped with multi-ratio current transformers for the purpose of metering and protection. The recloser shall be provided with multifunction meter to read the following parameters: ampere, voltage, KW demand, KVAR, KWH and Power factor. The protection C.T. shall cover the entire protection range of the device. Protection setting through the full range of the device should be programmable without changing the CTs. Minimum ground fault current at 33 KV bus is approximately 150 amps. So the time current characteristic curve for ground tripping must be rational with the phase current tripping and minimum line to ground fault current. The minimum phase or ground trip currents should be changeable externally in the control cubicle or in the electronic panel or by changing the CT ratio. CT's should be provided for the purposes of both metering and protection. Provision should be made for measuring phase load current and should be read from the 4 line LCD display in the control cubicle or from an electronic meter. Meter scale should be provided for lower range phase ampere measurement. These CTs shall meet the applicable requirements of the latest revision of ANSI C57.13 and shall have relay accuracy of class C100.

4.4 Bidder should supply minimum three (3) copies of catalogue showing the exploded view of each type(s) of equipment along with part numbers so that each component can be identified. With each equipment one (1) copy of the operational manual should also be supplied.

4.5 Lock washers - all electrical connections, bushing mounting bolts and cover attachment bolts require lock washers. Lock washers shall be fabricated from materials that comply with the requirements of ANSI B18.21.1.

4.6 Substation Mounting Frame - The three-phase reclosers shall be mounted in a height-adjusting mounting frame to maintain a minimum ground distance of two hundred seventy five centimetres (9') to live parts and two hundred forty four centimetres (8') to any bushing porcelain above the concrete foundation. The mounting frame shall be made of steel to provide a sturdy, stable foundation for the recloser and be complete with provisions to secure the frame to the recloser concrete foundation plus secure the recloser to the mounting frame. The substation mounting frame shall have provisions to mount a recloser tank lowering windless compatible with substation mounting frame. For grounding purposes, all recloser mounting frames are to be drilled for two 1.42875 centimetres (9/16") round holes spaced on 4.445 centimetres (1.75") centres and located on opposite corner legs 30.48 centimetres (1') from the bottom of the frame. The mounting frame shall be hot dipped galvanized after fabrication.

5. **OPERATION CHARACTERISTICS:**

5.1 Operation - An REB Class III recloser shall be capable of sensing and interrupting fault current within the range of its maximum trip and maximum interrupting ratings. For permanent faults, the recloser shall lock-out after two, three or four operations depending on the control settings. Operation of a typical class III recloser with four operations (two fast and two slow) to lock-out is illustrated in Figure 1.

5.2 Co-ordination - Class III 33KV recloser shall have dual time-current capability. They shall have capability for field adjustment and selection of alternate curves (Standard inverse, very inverse or extremely inverse) in order to properly co-ordinate with the source relay characteristics curve sequence of operation and number of operations to lock-out. The recloser shall be capable of field adjustment to co-ordinate with the following:

a. The sub-station is 33:11/6.35 KV delta-Wye. The 33 KV automatic circuit recloser will be installed in between the 33 KV source circuit breaker and the 100E standard speed fuse at the 33 KV bus of REB substation. Class III 33 KV recloser must be adjustable to trip to lock out later than the maximum clearing time of the indicated power fuses and faster than the damage curve of 33 KV side conductor as well as the 33/11 KV power transformer (5 or 10 MVA, with about 6% impedance). Normally Merlin, Wolf and or 4/0 ACSR conductors are used in the 33 KV side of the substation.

c. 33KV, 3 phase reclosers are installed in the source side before the power fuse. The class III-11KV reclosers must trip to lock-out before the 33KV fuse and 33KV, 3 phase recloser. 33KV, 3 phase class III recloser's time-current characteristic curves are shown in Figure 2.

d. For bid evaluation, each supplier must furnish drawings showing proposed recloser time-current characteristics of operating curves which will co-ordinate between limits as shown in Figure 2.

5.3 Number of Operations - Factory settings for phase and ground trip operating sequences shall be as given in Table I. Class III reclosers shall have provision for field adjustment to provide the following options on either phase or ground trip:

a. Number of fast operations - one, two, three or four.

b. Total operations to lock-out, two, three or four.

c. Non-reclosing features - operable by external lever or provision in electronic control to provide one operation to lockout without reclosing.

d. The bidder shall submit the software package to set the complete set.

5.4 Reclosing Interval - Adjustable from 0.6 to 2.5 seconds with a 0.1 second resolution.

5.5 Reset Time; The reset time of the reclosers should be variable from o second to 10 seconds

6. TESTS:

6.1 Design tests shall be carried out by the manufacturer from an internationally recognized independent testing laboratory in accordance with Section 6 of ANSI/IEEE C37.60.& relevant IEC standards. The manufacturer shall submit certified copies of design test reports along with bid document. Design tests on REB class III reclosers and control panels shall include, but not be limited to, the following:

(1) Recloser

a. Impulse withstand voltage test
b. Low frequency withstand voltage tests.
c. Interruption tests.
d. Making current tests.
e. Minimum tripping current tests.
f. Radio influence voltage tests.
g. Temperature rises test.
h. Time-current test.
i. Mechanical operations test.
j. Transformer magnetizing current interruption test
k. Simulated surge arrester operation test.

II) Control Panel

a. Electrostatic Discharge immunity test
b. Fast transient/Burst immunity test
c. Radiated radio frequency immunity test
d. Surge withstand capability immunity test
e. Cold, damp heat, cyclic, dry heat test
f. Encloser Protection
g. Vibration test
h. Dielectric Withstand test
i. Impulse test

6.2 Production tests shall be carried out by the manufacturer in conformance with the latest revision of Section 7 of ANSI/IEEE C37.60. The manufacturer shall submit to REB certified copies of production test reports. Production tests on REB Class III reclosers shall include, but not be limited to, the following:

a. Calibration test.

b. Control, secondary devices and accessory devices tests.

c. Dielectric withstands test - 1 minute dry, low frequency.

d. No load operating test.

e. Mechanical operations test.

7. Warranty:

The Contractor shall warrant that the ACR furnished have conformed to this specification. The warranty shall state that if, within three(3)years from the date of delivery in case of EXW contracts & from the date of arrival at the designated port of entry in case of CFR contracts , a ACR is found to have defects in workmanship or material (or fails in service due to such defects) the Contractor shall repair or replace such defective parts (and other parts damaged as a result) as soon as possible, free of charge

8. SPARE PARTS:

The bidder shall include separately with his offer a list of manufacturer's recommended spare parts for two (2) years operation per unit with CIF price quoted for each of the items. The cost of spare parts will not be considered in the tender evaluation. The purchaser reserves the option of purchasing any or all of the spare parts listed.

9. **OTHER STANDARDS:**

The dimensional and performance requirements of the REB Class III recloser based on other internationally recognized standards are acceptable only if the requirements of such standards are equivalent to or exceed the requirements quoted in this document. As co-ordination with other reclosers and sectionalizing devices is critical for protection of the REB distribution systems, the standards in this document covering operating characteristics shall be strictly adhered to.

10. **PACKAGING:**

10.1 Refer to "Export Packing" Shipping Box Design Number 2 for shipping box requirements.

10.2 Reclosers must be securely fastened within the shipping box to prevent damage during shipment and handling. Particular attention shall be given to the protection of the porcelain bushings and bushing of the current transformers.

11. **INFORMATION REQUIRED WITH TENDER:**

The Bidder shall furnish the following type test reports from an independent testing laboratory with his bid proposal for each size and Class of recloser included in the proposal:

a. Rated symmetrical interrupting current.

b. Rated impulse withstands voltage.

c. Low frequency withstand voltage at:
60 seconds dry
10 seconds, wet

d. Time current curves of the recloser phase and ground trip plotted to the same scale as the curves in Figure 2 of this standard. The curves shall be based on tests conducted at 25C.

12. **BIBLIOGRAPHY OF REFERENCED STANDARDS: Latest Edition**

1: ANSI/IEEE C37.60, American National Standard for Overhead, Pad-Mounted, Dry Vault and Submersible Automatic Circuit Reclosers and Fault Interrupters for AC Systems.

2: ASTM D3487, Standard Specification for Mineral Insulating Oil used in Electrical Apparatus.

3: ANSIC57.13 American National Standard for Instrument Transformers.

4: ANSI B18.21.1, American National Standard for lock washers.

TABLE I

CURRENT RATINGS
AND
OPERATING SEQUENCES

REB STANDARD - CLASS III

33 KV THREE PHASE RECLOSERS

Item #	Rated symmetrical interrupting capacity	Phase trip		Ground trip	
		Minimum trip Amps variable from	Number of fast curve	Operations slow curve	Minimum trip Amps variable from
H-7.400B	12,000	100 to 400	0 - 4	0 – 4	10 to 400
H-7.630B	12,000	100 to 630	0 – 4	0 – 4	10 to 630
H-7.800B	16,000	100 to 800	0 – 4	0 - 4	10 to 800

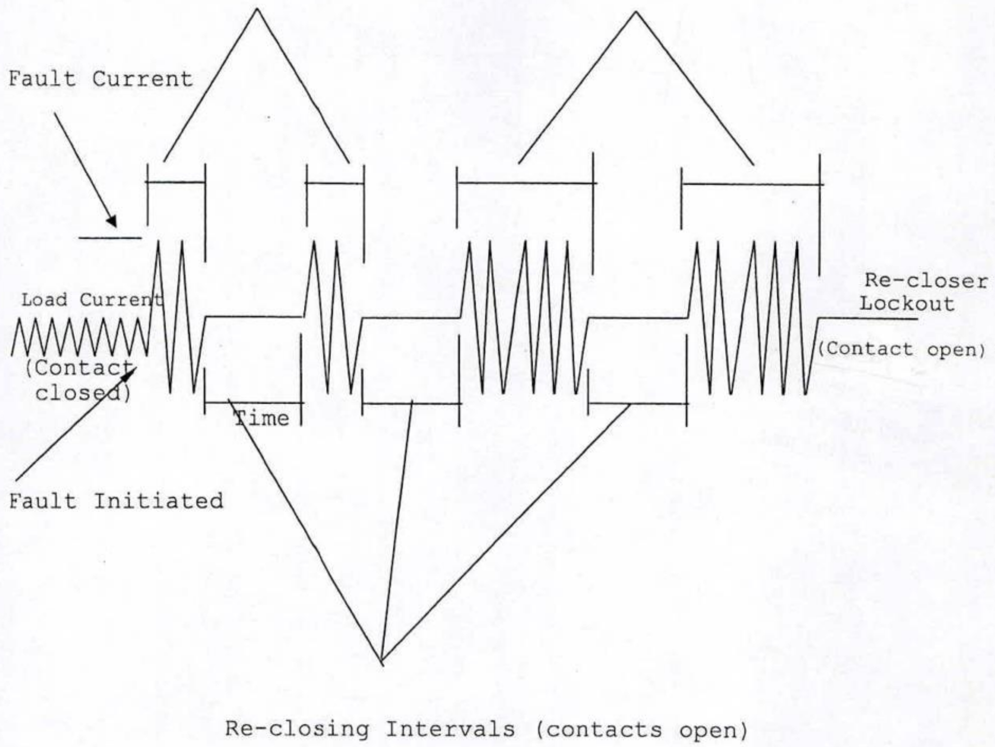

Figure 1
OPERATING SEQUENCES
FOUR OPERATIONS TO LOCKOUT

সহকারী প্রকৌশলী
বাপিএসএস, বাপবিবো, ঢাকা।

REB ITEMS H-7.400B Through 7.800B

(মোঃ সদর-ঈ-রাফাত)
সহকারী প্রকৌশলী
পিএসএস, বাপবিবো, ঢাকা।

PUBLICATION 307B-2010
RURAL ELECTRIFICATION BOARD (REB)
PEOPLES REPUBLIC OF BANGLADESH
STANDARD FOR CLASS III
11 KV THREE PHASE SOLID DI-ELECTRIC
AUTOMATIC CIRCUIT RECLOSERS

1. **SCOPE:**

This standard covers the electrical, mechanical and performance requirements for three phase, automatic circuit reclosers for use on the REB overhead distribution system and in REB substations.

2. **GENERAL:**

Reclosers furnished to REB specifications shall conform in all respects to the requirements of this standard. These reclosers shall be suitable for use in a tropical climate with high relative humidity up to 95%. The text, tables, figures and performances to other standards supplement each other and are considered part of this standard.

3. **RATINGS:**

3.1 Service Conditions:

a. The required operating temperature range shall be between-10C to + 65^0 C.
b. Altitude - up to 1,000 meters.

3.2 Rated Voltage

a. Nominal system voltage-11/6.35 KV Grd. Y.
b. Rated maximum voltage - 15,000 volts.
c. Low frequency withstand voltage, RMS
i) Dry, 60 seconds-50 KV
ii) Wet, 10 seconds-45 KV

3.3 Rated frequency - 50 Hertz.

3.4 Rated minimum current shall be as shown in Table I.

3.5 Rated minimum tripping current shall be as shown in Table I with a tolerance of ±percent.

3.6 Rated symmetrical interrupting current shall be as in Table I with asymmetrical relationships as defined in ANSI/IEEE C37.60.

3.7 Rated symmetrical making current shall be the same value as the rated symmetrical interrupting current, with asymmetrical relationships as defined in ANSI/IEEE C37.60.

3.8 Rated minimum Impulse Withstand Voltage - 110 KV.

4. DESIGN

Design of the REB class III recloser shall be in accordance with ANSI/IEEE C37.60. The recloser shall be designed to provide self-contained distribution circuit protection with series trip, magnetic actuators and electronic control double or single break contact structure and fault interruption under vacuum. The vacuum interrupters shall be in accordance with ANSI/IEEE C37.85. The recloser shall utilise vacuum interrupter as the interruption medium and solid dielectric as the insulation medium. The recloser shall be designed to sense line-to-ground, line-to-line or three-phase fault currents and interrupt all three phases of the distribution circuit simultaneously. The recloser shall be equipped with an electronic type ground trip. If the Electronic control cubicle of Reclosers or Recloser itself require an auxiliary transformer, the manufacturer must provide the same at his cost. The recloser shall be provided with in-built auxiliary transformer of required rating to provide power supply to the control panel. The secondary voltage of the said PT must be 240V.

For 11KV solid dielectric ACR, terminal to terminal and lower terminal to ground/earth clearance shall not be less than 350 mm and 520 mm respectively. Terminal to terminal and lower terminal to ground/earth creepage distance shall not be less than 1180 mm and 671 mm respectively.

Design features are summarised below:

4.1 The Recloser head shall be fabricated of corrosion resistant materials or provided with an impact and corrosion resistant finish and should also be suitable for storage in uncovered areas and shall be provided with the following features:

a: Universal clamp type terminals which will accept #6 AWG to 350 MCM copper or aluminium conductors either horizontally or vertically. In place of universal clamp, insulated and water blocked cable tails of proper voltage and current rating is acceptable.

b: Silicon bushings with bushing clamps bolted to head with bird (guard) protection or epoxy bushing and EPDM rubber bushing boots to provgide a fully insulated bushing arrangement.

c: Lifting provision is required to lift the entire recloser so positioned that the recloser will remain level when lifted.

d: Manual Operating Provision: Recloser shall be provided with a manual operating lever, which is suitable for operating with a hot stick or other suitable manual operating means.

e: Non-reclosing lever/feature to set recloser to lock out after one operation.

f: Counters: An operation counter shall be provided to indicate the total number of operations of the reclosers as well as the number of fault operations for each type of fault. The counter shall be visible with recloser in service.

g: Attachment clamps and an "0-ring" seal to fix tank to head in such a manner as to provide a positive, moisture-proof seal.

h: Closing tool port for manually closing de-energized recloser, Provisions should be made for manually closing de-energized recloser without un-tanking it. A manual closing tool shall be provided with each recloser. Alternatively closing and tripping by LV solenoid and control cubicle is acceptable.

i: Nameplate(s) listing factory ratings and settings of the recloser, with provisions for changing of nameplate data when field changes are made affecting operating characteristics of the recloser. Nameplate information shall include, but not be limited to the following:

Manufacturer's identification

Contract number and IFB number

Recloser type.

Recloser serial number.

Rated maximum voltage.

Rated minimum continuous current.

Rated minimum tripping current.

Rated minimum symmetrical interrupting current.

Rated minimum impulse withstand voltage.

Number of fast operations & also identification of Operational Curve..

Number of slow operations & also identification of Operational Curve.

Total operations to lock-out.

Designation of operating curves.

BIL Rating.

Nominal system voltage.

Year of manufacture.

4.2 The recloser tank shall be fabricated of corrosion resistant steel, and provided with an impact and corrosion resistant finish and should also be suitable for storage in uncovered area and provided with a mating surface for the head so that leaks will not occur and that the recloser will remain mechanically operable at the maximum operating pressure generated by normal operation of the recloser. The tank shall have the following features:

a. Mounting lugs for mounting the recloser in a substation frame or cross arm mounting hanger.
b. Ground connector to accept #8 to #2/0 AWG copper or aluminum conductors.
c. The finish shall be a thermosetting impact and corrosion resisting, acrylic paint (ANSI Color #70) over inhibiting epoxy prime coat.

4.3 The electronic ground trip device shall be self-contained and independent. REB three phase recloser shall be equipped with multi-ratio current transformers for the purpose of metering and protection. The recloser shall be provided with multifunction meter to read the following parameters: ampere, voltage, KW demand, KVAR, KWH and Power factor. The protection C.T. shall cover the entire protection range of the device. Protection setting through the full range of the device should be programmable without changing the CTs. Minimum ground fault current at 11 KV bus is approximately 50 amps. So the time current characteristic curve for ground tripping must be rational with the phase current tripping and minimum line to ground fault current. The minimum phase or ground trip currents should be changeable externally in the control cubicle or in the electronic panel or by changing the CT ratio.

4.4 Bidder should supply minimum three (3) copies of catalogue showing the exploded view of each type(s) of equipment along with part numbers so that each component can be identified. With each equipment one (1) copy of the operational manual should also be supplied.

4.5 Lock washers - all electrical connections, bushing mounting bolts and cover attachment bolts require lock washers. Lock washers shall be fabricated from materials that comply with the requirements of ANSI B18.21.1.

4.6 Substation Mounting Frame - The three-phase reclosers shall be mounted in a height-adjusting mounting frame to maintain a minimum ground distance of two hundred seventy five centimetres (9') to live parts and two hundred forty four centimetres (8') to any bushing porcelain above the concrete foundation. The mounting frame shall be made of steel to provide a sturdy, stable foundation for the recloser and be complete with provisions to secure the frame to the recloser concrete foundation plus secure the recloser to the mounting frame. The substation mounting frame shall have provisions to mount a recloser tank lowering windless compatible with substation mounting frame. For grounding purposes, all recloser mounting frames are to be drilled for two 1.42875 centimetres (9/16") round holes spaced on 4.445 centimetres (1.75") centres and located on opposite corner legs 30.48 centimetres (1') from the bottom of the frame. The mounting frame shall be hot dipped galvanized after fabrication.

5. OPERATION CHARACTERISTICS:

5.1 Operation - An REB Class III recloser shall be capable of sensing and interrupting fault current within the range of its maximum trip and maximum interrupting ratings. For permanent faults, the recloser shall lock-out after two, three or four operations depending on the control settings. Operation of a typical class III recloser with four operations to lock-out is illustrated in Figure 1.

5.2 Co-ordination - Class III 11KV recloser shall have dual time-current capability. They shall have capability for field adjustment and selection of alternate curves (Standard inverse, very inverse or extremely inverse) in order to properly co-ordinate with the source relay characteristics curve sequence of operation and number of operations to lock-out. The recloser shall be capable of field adjustment to co-ordinate with the (1) single phase 6.35KV line recloser, (2) 33KV power fuse and (3) 33 KV 3 phase recloser. Beginning at the 33 KV source in the direction of the 11 KV load, the arrangements is as follows: 33 KV 3 phase class III recloser, 33KV power fuse, substation power transformer, class III-11KV recloser and single phase 11KV line recloser.

a. Load side (11KV) - In the REB distribution system there are different sizes of single-phase 6.35 KV line reclosers from different manufacturers. The 3 phase class III 11 KV REB 3 phase reclosers must trip and lock-out later than single phase line reclosers, which may be set for two fast curve and two delayed curve operations to lock-out. Between the fast and slow operations there must be sufficient time delay to allow for operation of fuses.

The time-current characteristic operating curves of the single phase line reclosers fuses are shown in Figure 2.

b. Source side (33KV) - the substation is 33:11/6.35 KV delta-wye. 100E standard speed fuses are installed in the 33KV source side of the substation. Class III REB 11KV 3 phase reclosers must trip to lock-out faster than the minimum melting time of the indicated power fuse as shown in Figure 2.

c. 33KV, 3 phase reclosers are installed in the source side before the power fuse. The class III-11KV reclosers must trip to lock-out before the 33KV fuse and 33KV, 3 phase recloser. 33KV, 3 phase class III recloser's time-current characteristic curves are shown in Figure 2.

d. For bid evaluation, each supplier must furnish drawings showing proposed recloser time-current characteristics of operating curves which will co-ordinate between limits as shown in Figure 2.

5.3 Number of Operations - Factory settings for phase and ground trip operating sequences shall be as given in Table I. Class III reclosers shall have provision for field adjustment to provide the following options on either phase or ground trip:

a. Number of fast operations - one, two, three or four.

b. Total operations to lock-out, two, three or four.

c. Non-reclosing features - operable by external lever or provision in electronic control to provide one operation to lockout without reclosing.

d. The bidder shall submit the software package to set the complete set.

5.4 Reclosing Interval - Adjustable from 0.6 to 2.5 seconds with a 0.1 second resolution.

5.5 Reset Time; The reset time of the reclosers should be variable from o second to 10 seconds.

6. TESTS:

6.1 Design tests shall be carried out by the manufacture from an internationally recognized independent testing laboratory in accordance with Section 6 of ANSI/IEEE C37.60.& relevant IEC standards. The manufacturer shall submit certified copies of design test reports along with bid document. Design tests on REB class III reclosers and control panels shall include, but not be limited to, the following:

1) Recloser

a. Impulse withstand voltage test
b. Low frequency withstand voltage tests.
c. Interruption tests.
d. Making current tests.
e. Minimum tripping current tests.
f. Radio influence voltage tests.
g. Temperature rises test.
h. Time-current test.
i. Mechanical operations test.
j. Transformer magnetizing current interruption test.
k. Simulated surge arrester operation test.

II) Control Panel

a. Electrostatic Discharge immunity test
b. Fast transient/Burst immunity test
c. Radiated radio frequency immunity test
d. Surge withstand capability immunity test
e. Cold, damp heat, cyclic, dry heat test
f. Encloser Protection
g. Vibration test
h. Dielectric Withstand test
i. Impulse test

6.2 Production tests shall be carried out by the manufacturer in conformance with the latest revision of Section 7 of ANSI/IEEE C37.60. The manufacturer shall submit to REB certified copies of production test reports. Production tests on REB Class III reclosers shall include, but not be limited to, the following:

a. Calibration test.

b. Control, secondary devices and accessory devices tests.

c. Dielectric withstands test - 1 minute dry, low frequency.

d. No load operating test.

e. Mechanical operations test.

7. **Warranty:**

The Contractor shall warrant that the ACR furnished have conformed to this specification. The warranty shall state that if, within three(3)years from the date of delivery in case of EXW contracts & from the date of arrival at the designated port of entry in case of CFR contracts , a ACR is found to have defects in workmanship or material (or fails in service due to such defects) the Contractor shall repair or replace such defective parts (and other parts damaged as a result) as soon as possible, free of charge

8. **OTHER STANDARDS:**

The dimensional and performance requirements of the REB Class III recloser based on other internationally recognized standards are acceptable only if the requirements of such standards are equivalent to or exceed the requirements quoted in this document. As co-ordination with other reclosers and sectionalizing devices is critical for protection of the REB distribution systems, the standards in this document covering operating characteristics shall be strictly adhered to.

9. **PACKING:**

9.1 Refer to "Export Packing" Shipping Box Design Number 2 for shipping box requirements.

9.2 Reclosers must be securely fastened within the shipping box to prevent damage during shipment and handling. Particular attention shall be given to the protection of the porcelain bushings and bushing of the current transformers.

10. **INFORMATION REQUIRED WITH TENDER:**

The Bidder shall furnish the following type test reports from an independent testing laboratory with his bid proposal for each size and Class of recloser included in the proposal:

a. Rated symmetrical interrupting current.

b. Rated impulse withstands voltage.

c. Low frequency withstand voltage at:
60 seconds dry
10 seconds, wet

d. Time current curves of the recloser phase and ground trip plotted to the same scale as the curves in Figure 2 of this standard. The curves shall be based on tests conducted at 25C.

11. **SPARE PARTS:**

The Bidder shall include separately with his offer a list of manufacturer's recommended spare parts for 2 (two) years operation per unit with CIF price quoted for each of the item. The cost of spare parts will not be considered in the tender evaluation. The purchaser reserves the option of purchasing any or all of the spare parts listed.

12. **BIBLIOGRAPHY OF REFERENCED STANDARDS: Latest Edition**

1: ANSI/IEEE C37.60, American National Standard for Automatic, Circuit Reclosers For Alternating Current system.

2: ASTM D3487, Standard Specification for Mineral Insulating Oil used in Electrical Apparatus.

3: ANSIC57.13 American National Standard for Instrument Transformers.

4: ANSI B18.21.1, American National Standard for lock washers.

5: C37.61 American National Standard guide for the application operatipn & maintenance of Automatic Circuit Recloser.

TABLE I

CURRENT RATINGS
AND
OPERATING SEQUENCES

REB STANDARD - CLASS III

11 KV THREE PHASE RECLOSERS

Item #	Rated symmetrical interrupting capacity	Phase trip		Ground trip	
		Minimum trip Amps variable from	Number of fast curve	Operations slow curve	Minimum trip Amps variable from
H-8.200	12,000	100 to 200	0 - 4	0 – 4	10 to 200
H-8.400	12,000	100 to 400	0 – 4	0 – 4	10 to 400
H-8.500	16,000	100 to 500	0 – 4	0 - 4	10 to 500

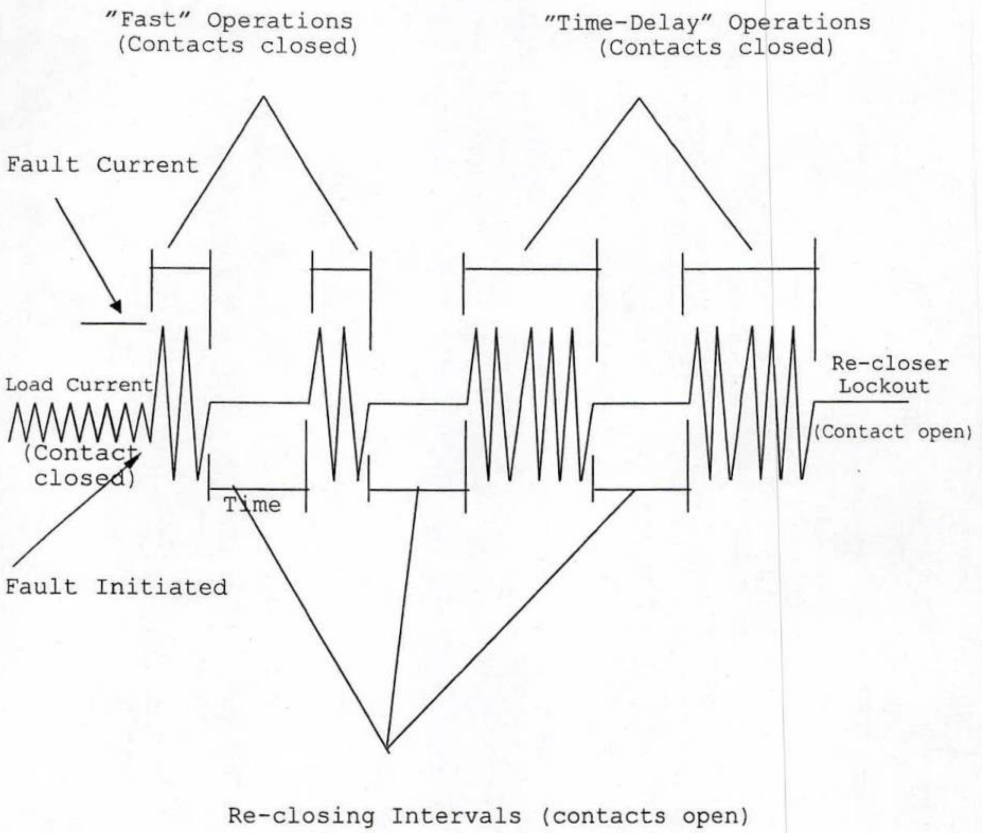

Figure 1
OPERATING SEQUENCES
FOUR OPERATIONS TO LOCKOUT

(মোঃ সদর-উ-রাফাত)
সহকারী প্রকৌশলী
এমপিএসএস, বাপবিবো, ঢাকা।

SPECIFICATION SUBMISSION AND COMPLIANCE SHEET(Form PG3-5) 1 of 2

(Attachment to the Tender submission sheet)
(To be filled by the Tenderer)

Bid Package No.: MCEP/BREB/KD-G-20
Sub-Package No.: Lot-1
Item : 11 kv ACR (HS-8.400B)

Item No.	Technical Particulars	REB Specification	Guaranteed Specification
H-8.400B (400 Amps)	Manufacturer's Name	To be mentioned	
	1. Design standard	ANSI C37.60	
	2. Catalog no. of proposed item.	To be mentioned	
	3. Country of origin.	To be mentioned	
	4. Installation Place	Outdoor, at 11 kv Feederof 33/11 Substations.	
	5. Environment of Installation Place	Temperatue- upto 65^{0}C, Humidity up to 95% & Altitude upto 1,000 meters	
	6. Circuit System	3 Phase, 11 KV, 50 Hz.	
	7. Nominal System Voltage/Rated Max voltage	11 KV/15 KV	
	8. Interrupting Medium	Vacuum, Vacuum Interrupters shall be in accordance with ANSI/IEEE C37.85	
	9. Control System	Electronic	
	10. Ground Trip Mechanism	Electronic	
	11. Insulating medium	Solid di-electric	
	12. BIL	110 KV	
	13. Low frequency voltage withstand	Dry-60 Sec : 50 KV	
		Wet- 10 Sec: 45KV	
	14. Fault type	The recloser shall be designd to sense line-to-ground, line-to-line or three phase fault current and shall interrupt all three phase of the distribution circuit simultaneously.	
	15. Ground/earth clearence (minimum)	350/520mm	
	16. Ground/earth creepage distence(min$^{m)}$	1180/671 mm	
	17. Rated Symetrical Interrupting current	12,000 Amps	
	18. Phase trip continuous current	400 Amps	
	19. Rated Min. phase tripping current	100-400 Amps	
	20. Rated Min. ground tripping current	10-400 Amps	
	21. Break Contact Structure	single break or double break	
	22. Number of fast curves	0 to 4	
	23. Number of slow operation	0 to 4	
	24. Recloser head	Corrosion resistance Material	
	25. Recloser Tank	Corrosion resisting Steel	
	26. Connector	# 6 AWG to 350 MCM Cu or AL	
	27. Recloser Bushing	Silicon bushing with bushing clamps bolted to head with bird guard or Epoxy bushing and EPDM rubber bushing boots to provide a fully insulated bushing arrangement.	
	28. Lifting Provision	Shall be provided to lift with level	
	29. Manual Operation Provision	Shall be provided with manual operating lever	
	30. Lockout Provision	Non reclosing lever/feature to set recloser to lockout after one operation.	
	31. Operation Counter	Shall indicate total operation	

Item No.	Technical Particulars	REB Specification	Guaranteed Specification
H-8.400B (400 Amps)	32. Manual Closing tool	A manual closing tool shall be provided with each recloser.	
	33. Name Plate information	Shall be provided as stated in the specification.	
	34. Current Transformer	Recloser shall be equipped with multi-ratio CT for metering & protection. CT shall cover the entire protection range.	
	35. Metering Provision	Recloser shall be provided with 3 phase Multifunction meter for reading A, V, KW, KVAR, KWH R & P.F.	
	36. Protection Settings	Current, time and related operational settings shall be programmable through the self-contained built-in keyboard & display of the Control cubical of the ACR itself. Protection setting through the full range of the device should be programmable without changing the CTs.	
	37. Co-ordination Capability (TCC Curve)	Class III 11 KV reclosure shall have dual time current capability for field adjustment and selection of alternate curves of Standard inverse, Very inverse or extremely inverse.	
		The recloser shall be capable of to co-ordinate with 6.35 KV line reclosure, 33 Kv power fuses, 33 KV 3 pase reclosure and must be adjustable to trip to lock out later than the single phase line reclosures, which may be set for two fast curve and two delayed curve operations to lockout. Between the fast and slow operations there may be sufficient time delay to allow for operation of fuses.	
	38. Factory Setting for Phase and Ground trip operating sequences	Shall be as given in Table 1 of specification	
	39. Provision for field adjustment	No. of fast operation: - one, two, three or four.	
		Total operation to lock-out :-two, three or four	
		Non-reclosing features-Operable by external lever or provision in electronic control to provide one operation to lock-out without reclosing.	
	40. Reclosing Time Interval	Adjustable 0.6 to 2.5 Sec with 0.1 sec. resolution	
	41. Reset Time	Variable from 0 sec. to 10 seconds.	
	42. Prov. for different trip current setting	Without changing CTs	
	43. Substation mounting frame	Shall be provided	
	44. Auxilliary power supply	The reclosure shall be provided with auxiliary transformer, Battery & Charger of required rating to provide power supply to the control panel. The PT output must be 240 V.	
	45. List of spare parts	List of manufacturer's recommended spare parts for 2 years with price shall be submitted.	
	46. Design & Product Test	Different Tests as mentioned in the specification shall be carried out as per ANSI C37.60 & relevant standard from Internationally recognized independent testing laboratory and shall be submitted with the bid proposal.	
	47. Warranty	The bider shall warrant that the ACR furnished have conformed to REB specification. Within 3 Years from the date of delivery if the ACR is found to have defects, the bidder shall repair or replace the defective parts at free of cost.	
	48. Packing	As per shipping box design no. 2 of exhibit H-2 of tender document	

Note : The bidders are required to submit all certified Test Reports as per above description. In the event of non-submission of any Test Report, the bid may be rejected.

(মোঃ সদর-ঈ-রাফাত)
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(মোঃ রফিকুল ইসলাম)
উপ-পরিচালক (কারিগরী)
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SPECIFICATION SUBMISSION AND COMPLIANCE SHEET(Form PG3-5)

(To be filled by the Tenderer)

Package No.: MCEP/BREB/KD-G-20

Sub-Package No.: Lot-1

Item : 33 kv ACR (HS-7.800B)

Item No.	Technical Particulars	REB Specification	Guaranteed Specification
HS-7.800B	Manufacturer's Name	To be mentioned	
	1. Design standard	ANSI C37.60	
	2. Catalog no. of proposed item.	To be mentioned	
	3. Country of origin.	To be mentioned	
	4. Installation Place	At 33 KV Incomimg lines of REB S/S	
	5. Environment of Installation Place	Temperatue- upto 50^0C, Humidity up to 95% & Altitude upto 1,000 meters	
	6. Circuit System	3 Phase, 33 Kv, 50 Hz	
	7. Nominal System Voltage/Rated Max voltage	33 KV/38 KV	
	8. Interrupting Medium	Vacuum, Vacuum Interrupters shall be in accordance with ANSI/IEEE C37.85	
	9. Control System	Electronic	
	10. Ground Trip Mechanism	Electronic	
	11. Insulating medium	Solid-Dielectric	
	12. BIL	150 KV	
	13. Low frequency voltage withstand	Dry-60 Sec : 70 KV	
		Wet- 10 Sec: 60KV	
	14. Fault type	The recloser shall be designd to sense line-to-ground, line-to-line or three phase fault current and shall interrupt all three phase of the distribution circuit simultaneously.	
	15. Ground/earth clearence (minimum)	394/676mm	
	16. Ground/earth creepage distance (min^m)	1763/1215 mm	
	17. Rated Symetrical Interrupting current	16000 Amps	
	18. Rated Min. phase tripping current	100-800 Amps	
	19. Rated Min. ground tripping current	10-800 Amps	
	20. Break contact structure	single break or double break	
	21. Number of fast curve	0 to 4	
	22. Number of Slow operation	0 to 4	
	23. Recloser head	Corrosion resistance material	
	24. Recloser Tank	Corrosion resisting steel	
	25. Connector	# 6 AWG to 350 MCM Cu or AL	
	26. Recloser Bushing	Wet process porecelain with bushing clamps bolted to head with bird (guard) protection or Cyclosphatic bushing and EPDM rubber bushing boots to provide a fully bushing arrangement.	
	27. Lifting Provision	Shall be provided to lift with level	
	28. Manual Operation Provision	Shall be provided with manual operating lever	
	29. Lockout Provision	Non reclosing lever/feature to set recloser to lockout after one operation.	
	30. Operations Counter	Shall indicate total operation	
	31. Manual Closing tool	A manual closing tool shall be provided with each recloser.	
	32. Name Plate information	Shall be provided as stated in the specification.	
	33. Cable for control / meter box	10 meter long contol cable shall be provided	
	34. Current Transformer	Recloser shall be equipped with multi-ratio CT for metering & protection. CT shall cover the entire protection range.	
	35. Metering Provision	Recloser shall be provided with 3 phase Multifuction meter for reading A, V, KW, KVAR, KWH R & P.F.	

C-4\Naz.Evaluation HS-7.800B.xls

Item No.	Technical Particulars	REB Specification	Guaranteed Specification
HS-7.800B	36. Protection Settings	Current, time and related operational settings shall be programmable through the self-contained built-in keyboard & display of the Control cubical of the ACR itself. Protection setting through the full range of the device should be programmable without changing the CTs.	
	37. Co-ordination Capability (TCC Curve)	Class III 33 KV reclosure shall have dual time current capability for field adjustment and selection of alternate curves of Standard inverse, Very inverse or extremely inverse.	
		The recloser shall be capable of to co-ordinate with 100E/125E/175E standard speed power fuse at 33 KB Bus and must be adjustable to trip to lock out later than the maximum clearing time of the indicated power fuse and faster than the damage curve of 33 Kv side conductor as well as the 33/11 Power transformer of the S/S.	
	38. Factory Setting for Phase and Ground trip operating sequences	Shall be as given in Table 1 of specification	
	39. Provision for field adjustment	No. of fast operation: - one, two, three or four.	
		Total operation to lock-out :-two, three or four	
		Non-reclosing features-Operable by external lever or provision in electronic control to provide one operation to lock-out without reclosing.	
	40. Reclosing Time Interval	Adjustable 0.6 to 2.5 Sec with 0.1 sec. resolution	
	41. Reset Time	Variable from 20 sec. to 120 seconds.	
	42. Prov. for different trip current setting	Without changing CTs	
	43. Pole mounting hanger:	Pole mounting hanger shall be provided.	
	44. Auxilliary power supply	The reclosure shall be provided with auxiliary transformer, Battery & Charger of required rating to provide power supply to the control panel. The PT output must be 240 V	
	45. List of spare parts	List of manufacturer's recommended spare parts for 2 years with price shall be submitted.	
	46. Design & Product Test	Different Tests as mentioned in the specification shall be carried out as per ANSI C37.60 & relevant standard from Internationally recognized independent testing laboratory and shall be submitted with the bid proposal.	
	47. Warranty	The bider shall warrant that the ACR furnished have conformed to REB specification. Within 3 Years from the date of delivery if the ACR is found to have defects, the bidder shall repair or replace the defective parts at free of cost.	
	48. Packing	As per shipping box design no. 2 of exhibit H-2 of tender document	

Note:-The bidders are required to submit all certified Test Reports as per above description. In the event of non-submission of any Test Report, the bid may be rejected.

Signature of Tenderer:
Name:
In the capacity of:

(মোঃ সদর-ঈ-রাফাত)
সহকারী প্রকৌশলী
মপিএসএস, বাপবিবো, ঢাকা।

(মোঃ রফিকুল ইসলাম)
উপ-পরিচালক (কারিগরী)
এসপিএসএস পরিদপ্তর, বাপবিবো ঢাকা

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3. Drawings

There are drawings for each item included with the technical specifications described in sl. 2 of this section -6

4. Inspections and Tests

4.1 Pre-shipment/Pre-delivery Inspection	i) PSI: (a) **PSI Conducted by:** Pre-shipment Inspection will be carried out by the purchaser's nominated Inspection Team of 3 members and/or purchaser's appointed PSI agent. (b) **Time and Place of Inspection:** At least three weeks before the shipment **of** goods. Tests will be conducted in presence of the Purchaser's nominated inspection team/agent at the manufacturer's laboratory or any other recognized facility as accepted by the Purchaser. (c) **Process of Test:** Pre-shipment/Pre-delivery tests will be carried out as indicated in Section VI Technical Specifications and relevant standard. Team and/or PSI agent will randomly collect samples as indicated in Section VI Technical Specifications and following BREB existing rules. Each sample shall be identified and signed by the Inspection Team or PSI agent. The Inspection Team or PSI agent will inspect and witness the tests and will also conduct visual inspection of the physical conditions (identification mark, manufacturer's symbol and year, dimension, damage, deformity, finishing, etc.), quantity and packaging of the goods before shipment and delivery in accordance with GCC clause 26.2 **Cost:** All costs related to inspection & testing during pre-shipment inspection shall be borne by the supplier. However, costing related to travel, accommodation and subsidence of the purchaser's nominated inspectors/agent shall be borne by the purchaser. PSI agent fee (if applicable) will also be borne by the purchaser. If goods are not ready for inspection on due date and time as noticed by the supplier, the additional cost/PSI agent fee (if incurred) will be realized from the supplier. Notice**:** The supplier shall give a reasonable advance notice, including the place and time thereof to the Purchaser by e-mail/fax at least 45 (Forty Five) days in advance of the inspection. (d) **Shipment Clearance**: Purchaser will issue a shipment clearance to the supplier upon receipt of satisfactory PSI report within one week from the date of inspection.The supplier shall not ship the materials before obtaining shipment clearance from the BREB.

4.2 Post Landing/ Delivery Inspection	(a) The supplier shall inform the Purchaser immediately after arrival of the goods at Final destination. An inspection team nominated by the Purchaser shall perform the post landing inspection in presence of Supplier's representative (if they desire so). (b) The Inspection Team will collect samples and conduct tests following Technical Specifications/relevant standard as well as BREB existing rules. The Inspection Team will visually inspect the physical conditions (identification mark, manufacturer's symbol and year, dimension, damage, deformity etc.) and quantity of the goods delivered. If required delivered materials will be accepted after testing at BREB facilities and/or facilities within Bangladesh decided by Purchaser in presence of Supplier's representative (if they so desire). Further information on the tests to be carried out is provided in Section VI Technical Specifications. The cost of the post landing/delivery inspection will be borne by the Purchaser.

5. PACKING, MARKING AND DRAWINGS INSTRUCTIONS FOR BOX DESIGN

PACKING, MARKING AND DRAWINGS INSTRUCTIONS FOR BOX DESIGN NO. 1, 2 & 3

1.0 Packing Requirements:

All material and equipment shall be packed for export shipment to a tropical climate in wooden shipping boxes that are sufficiently durable to withstand numerous handing. The shipping boxes shall be strong enough to prevent loss from pilferage or damage sustained from stacking, shipping or handling. Workmanship in the manufacture of the wooden shipping boxes shall be of the highest standard and the material used shall be in accordance with the best commercial practices. It shall be the supplier's responsibility to provide packing acceptable to the insurance underwriters.

Sawn board lumber or plywood shall be used for the sheathing of the shipping boxes. All lumber used shall be new, well seasoned, sound, and free of splits, decay or an excessive number of knots. Plywood used shall be new, free of splits and decay and shall be exterior grade quality. Each shipping box shall contain only commodities of like items. These commodities shall be primary packaged in units of numerical quantities consistent with the manufacturer's normal packaging practices. However, the maximum numerical quantities of each item per package shall not exceed the limit as stated in the technical specifications for that particular item. The minimum protection to be afforded to commodities subject to damage from the elements or those items, which are packaged in cardboard cartons, is that a heavy plastic liner shall completely envelope the entire contents contained within the shipping box. Equipment shall be firmly bolted or strapped to the shipping box floor or otherwise braced, blocked or secured to the box to insure the security of the equipment during transit and handing. Special packing requirements that are stated in the technical specifications and the material and price schedule must be strictly observed.

2.0 Shipping Box Specifications:

There are three different shipping box designs (No.1, 2 & 3) that are to be used for packing equipment and material of shipment. The shipping box to be constructed by the supplier will be dependent upon what material or equipment the supplier is shipping. The supplier must conform to the requirements as set forth in the following written specifications and the conceptual design as pictorially represented in this document.

Deviation, if any, from the BREB specifications in the wooden packing of supplied materials, shall be recovered through imposing penalty as per BREB Board Decision.

2.1 Shipping Box Design Number 1:

This shipping box design shall be used for the packaging and shipping of the following commodities:

a. Line Hardware ("B" Items),
b. Conductor to be Shipped in Coils ("F" Items),
c. Conductor Accessories ("E" Items),
d. Underground/Submarine Power Cable Accessories ("F" /"DS" Items,
e. Sectionalizing Devices (except Group Operated Air Break Switches and Reclosers), ("H" Items),
f. Connectors ("I" items),
g. Meters and Metering Accessories ("J" Items),
h. Capacitors and Accessories ("K" Items),
i. Street Lights and Accessories ("L" Items),
j. Miscellaneous ("M" Items),
k. Test Instruments ("TI" Items),
l. Line Tools ("TL" Items).

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This box design will not be applicable for the items B-42, B-43, B-62, B-63, B-65, B-87, B-88, B-90, BS-41 & BS-41.1. These items will be delivered in bare bundle. The box sheathing shall be fabricated from exterior grade plywood with a minimum thickness of three-eighth of an inch (3/8"), or sawn board lumbers. The maximum width of the sawn boards shall be six inches (6"). The minimum thickness of the sawn boards utilized for the sides and roof of the box shall be five-eight of an inch (5/8"), whereas the floorboards shall have a minimum thickness of one inch (1"). The minimum dimensions of the skid runner and header shall be three inches (3") by four inches (4"). The skid runners shall be hardwood. The minimum dimensions of the frames and cross-braces shall be one inch (1") by four inches (4"). One-half inch (1/2") bolts shall be used to secure the header to the skid runner.

The box shall be strapped with four flat Steel/PVC strapping bands secured with crimped steel banding clips. Two steel/PVC bands shall be installed vertically and two steel/PVC bands shall be installed horizontally (lengthwise). The minimum width of steel bands shall be three-fourth of an inch (3/4"). The width of PVC bands shall be 15mm (+/-0.50 mm) and minimum thickness shall be 01 (one) mm. The nominal dimensions of Shipping Box Design Number 1 shall be three-fourth (3/4) of a meter in height, three-fourth (3/4) of a meter in width and one meter in length (Height=0.75M, Width=0.75M, Length=1.0M). Note that these are nominal dimensions and that the Supplier may slightly vary these dimensions to better accommodate the packaged commodities.

In addition to, only for local bidder, in case of line hardware (B items) supply of Wooden Box (Wooden Box Design No 1) can be omitted. Instead of Wooden Box, they can supply line hardware packaged by Gunny bag/Jute bag. Gunny bag/ Jute bag must be double layer, unused and durable. Each package shall be clearly marked with the manufacturer's name, BREB item number & Quantity.

2.2 Shipping Box Design Number 2:

This shipping box design shall be used for the packaging and shipping of the following commodities:

a. Transformers ("G" Items),
b. Automatic Circuit Reclosers ("H" Items),
c. Group Operated Air Break Switches ("H" Items),
d. Automatic Voltage Regulators ("K" Items),
e. Static Converters ("M" Items).

The material and fabrication requirements for Shipping Box Design Number 2 shall be the same as the material and fabrication requirements of Shipping Box Design Number 1 except for the following differences: (i) cross-bracing is not required; (ii) only three steel/PVC bands are required, two bands installed vertically and one band installed horizontally (lengthwise); and (iii) there are no nominal dimensions. The minimum width of steel bands shall be three-fourth of an inch (3/4"). The width of PVC bands shall be 15mm (+/-0.50 mm) and minimum thickness shall be 01 (one) mm. The size of the shipping boxes will be determined by the maximum number of pieces of equipment allowable per box. The maximum number of pieces of equipment allowable per box is as follows:

a. Transformers, 15 kVA and below ---2 per box,
b. Transformers 25 kVA and above ---1 per box,
c. Automatic Circuit Reclosers, single phase ---2 per box,
d. Automatic Circuit Reclosers, three phase ---1 per box,
e. Group Operated Air Break Switches ---1 per box,
f. Automatic Voltage Regulators ---1 per box,
g. Static Converters ---1 per box.

2.3 Shipping Box Design Number 3:

This shipping box design shall be used for the packaging and shipping of insulators ("C" items). The box shall be fabricated from chemically treated sawn board lumber with minimum sizes as given in table-1. All frames shall be installed only on the outside of the box. Wooden separators shall be used to individually partition both horizontally and vertically, each insulator. The box shall be strapped with two flat Steel/PVC strapping bands, vertically installed. The minimum width of steel bands shall be one-half inch (1/2"). The width of PVC bands shall be 15mm (+/-0.50 mm) and minimum thickness shall be 01 mm. Item wise number of insulator per box is shown on the table-1

3.0 Export Marking:

All packages shall be identified with the Export Marking Instruction as set forth in the Delivery Schedule. In addition, the contents of the box, the net and gross weights and the Contract identification number shall be externally marked. All external package markings shall be legibly stenciled or painted with a quality marking substance that is both waterproof and durable. Chalk or crayon shall not be used. Packages must be numbered consecutively and no two packages delivered by the Supplier to each port of entry shall carry the same number regardless of the number of shipments. All steel banding clips used to secure the steel banding shall be legibly and indelibly marked with the manufacturer's name or symbol. Refer to detail of, Banding Clip Manufacturer's identification as depicted on the Shipping Box Design pictorial representation.

The words BREB BANGLADESH (GOB Fund) must be legibly written in English, using red capital letters at least four centimeters (4cm) in height, placed diagonally on four sides of all packages. In addition, all boxes shall be marked with one black and one orange strip depicted as "A" and "B" respectively on the painted corner detail identification of the Shipping Box Design pictorial representation.

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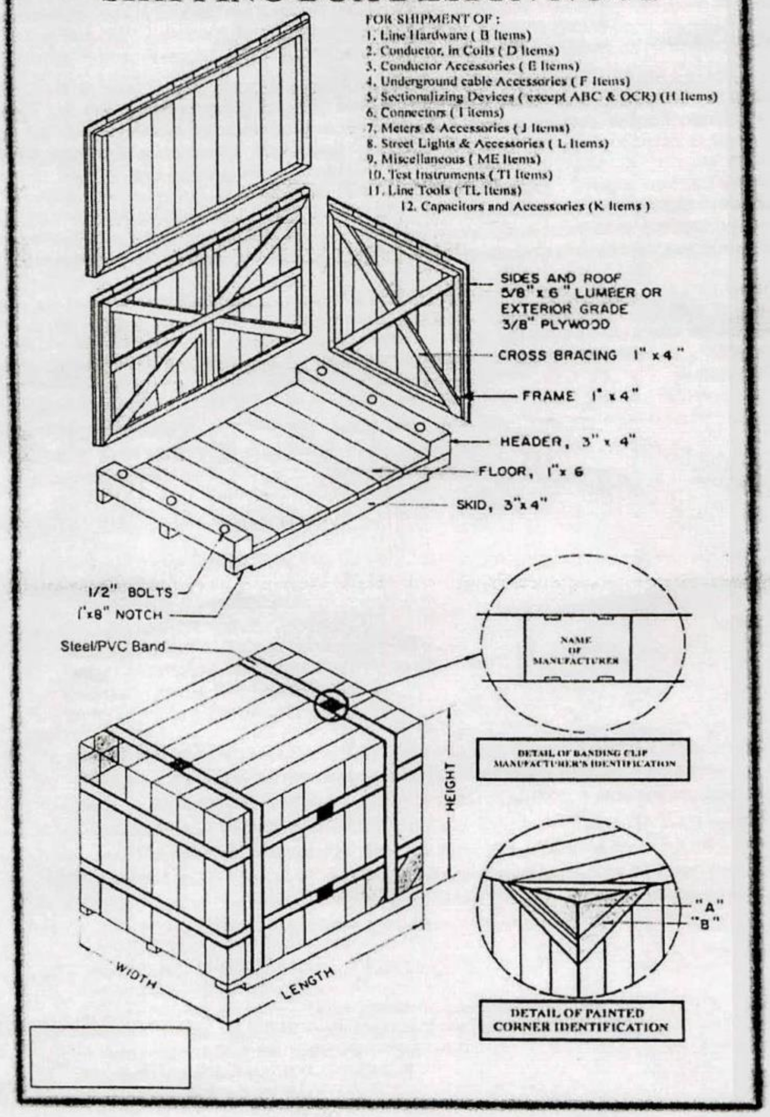
SHIPPING BOX DESIGN NO . 1



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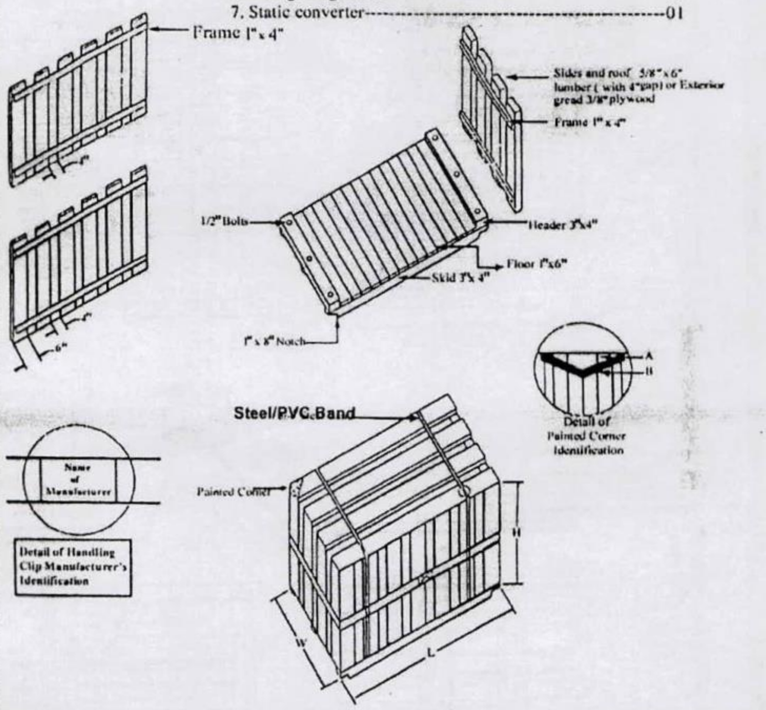
SHIPPING BOX DESIGN-2

FOR SHIPMENT OF EQUIPMENT	MAXIMUM PER BOX
1. Transformer 15KVA & below	02
2. Transformer 25KVA & below	01
3. Recloser, single phase	02
4. Recloser, three phase	01
5. Ground operated air break switch	01
6. Voltage Regulators	01
7. Static converter	01



Publication 491-2019, Revision 2, Date: April. 2019

SHIPPING BOX DESIGN ON.- 3 (Revised July 2000)

For Shipment of Insulator (C-Items)

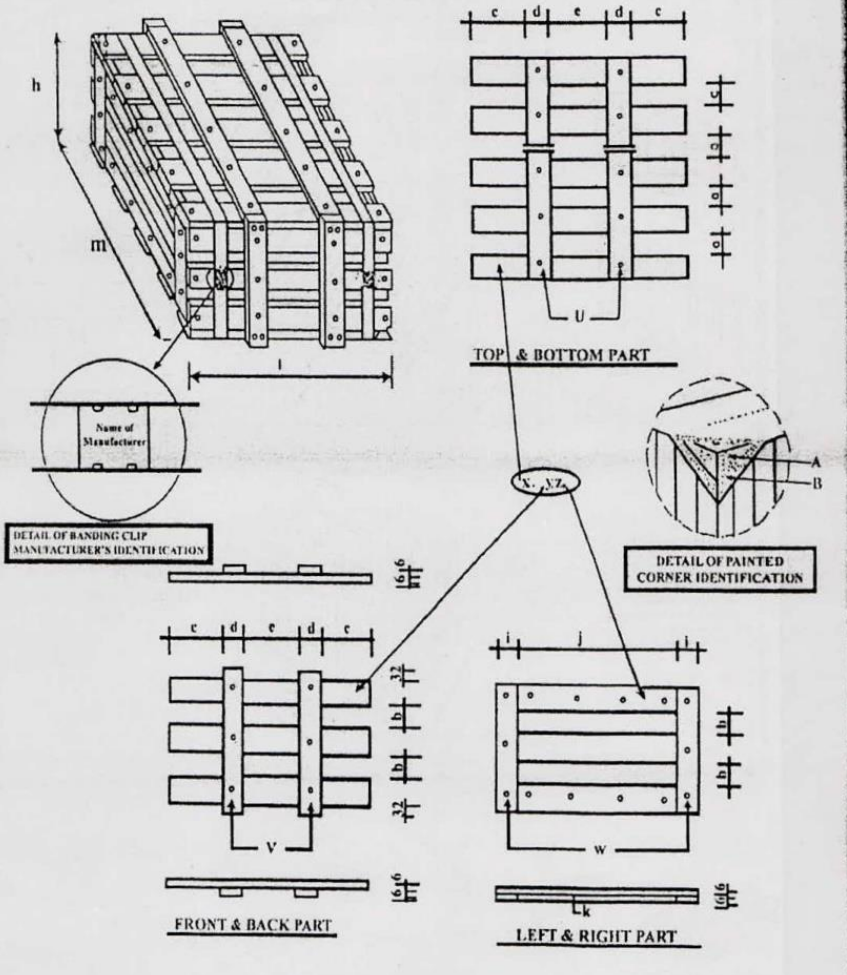


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Table-1

Item	Dimension of Insulator Box, Batten & Separator in mm					No. of Insulator per box
	C-1	C-2/C-3/C-4	C-5	C-10	C-11	
l x m x h	835 x 340 x 270	645 x 400 x 300	645 x 610 x 225	560 x 395 x 360	610 x 580 x 300	
x'	2 x 4 x 835 x 50 x 16	2 x 3 x 645 x 70 x 16	2 x 5 x 645 x 60 x 16	2 x 4 x 560 x 75 x 16	2 x 5 x 610 x 70 x 16	C-1=20 Pcs
y'	2 x 3 x 835 x 50 x 16	2 x 3 x 645 x 70 x 16	2 x 3 x 645 x 60 x 16	2 x 3 x 560 x 75 x 16	2 x 3 x 610 x 70 x 16	C-2=45 Pcs
z'	2 x 3 x 308 x 50 x 16	2 x 3 x 268 x 70 x 16	2 x 3 x 578 x 60 x 16	2 x 3 x 363 x 75 x 16	2 x 3 x 548 x 70 x 16	C-3=72 Pcs
k'	2 x 2 x 208 x 50 x 16	2 x 2 x 268 x 70 x 16	2 x 2 x 458 x 60 x 16	2 x 2 x 263 x 75 x 16	2 x 2 x 428 x 70 x 16	C-4=144 Pcs
u'	2 x 2 x 340 x 50 x 16	2 x 2 x 400 x 50 x 16	2 x 2 x 610 x 60 x 16	2 x 2 x 395 x 75 x 16	2 x 2 x 580 x 60 x 16	C-5=4 Pcs
v'	2 x 2 x 302 x 50 x 16	2 x 2 x 332 x 50 x 16	2 x 2 x 257 x 60 x 16	2 x 2 x 392 x 75 x 16	2 x 2 x 332 x 60 x 16	C-10=12 Pcs
w'	2 x 2 x 238 x 50 x 16	2 x 2 x 268 x 50 x 16	2 x 2 x 193 x 60 x 16	2 x 2 x 328 x 50 x 16	2 x 2 x 268 x 60 x 16	C-11=6 Pcs
a	4667	40	77.5	31 67	57.5	
b	44	29	65	51 5	29	
c	250	170	178	152	167	
d	50	50	60	75	60	
e	235	205	169	106	156	
i	50	50	60	50	60	
j	208	268	458	263	428	
Separator	i) 5 x 290 x 145 x 3.5 ii) 8 x 220 x 145 x 3.5 !) 2 x 755 x 100 x 3.5	i) N x 540 x 70 x 3.5 ii) N x 255 x 60 x 3.5	i) 4 x 575 x 60 x 18	I) 3 x 360 x 50 x 18 ii) 2 x 360 x 75 x 18 iii) 2 x 490 x 75 x 18	i) 3 x 548 x 60 x 18	

1) Allowable variations of l, m & h is +/- 5mm and accordingly of the length of x, y & z respectively.
2) Allowable variations of length of x, y, z, k, u, v, & w is +/- 5mm accordingly to the variations of l, m & h.
3) (') Marks means minimum breadth & width (thickness) are given in the above Table-1.
4) Minimum sizes of separators are given in the above Table-1 but according to the actual requirement higher sizes are acceptable.
5) "N" means "as required number".
6) All Framing shall be located on the outside of the box.
7) All insulators shall be separated both horizontally and vertically by wood partitions.
8) All wooden part shall be treated with chemical preservatives according to the requirement of REB.

(মোঃ জহিরুল ইসলাম)
প্রধান প্রকৌশলী (পওপ)
বাপবিবোর্ড, ঢাকা।

(এস,এম, জাফর সাদেক)
সদস্য (বিতরণ ও পরিচালন)

সদস্য (প্রশাসন), বাপবিবো

10-4-19

বাংলাদেশ পল্লী বিদ্যুতায়ন বোর্ড

(মোঃ ওমর ফারুক ভূঁইয়া)
সদস্য (সমিতি ব্যবস্থাপনা)

সদস্য (পরিকল্পনা ও উন্নয়ন)
বাংলাদেশ পল্লী বিদ্যুতায়ন বোর্ড

Publication 491-2019, Revision 2, Date: April. 2019

BANGLADESH RURAL ELECTRIFICATION BOARD, BANGLADESH
OFFICE OF THE ENVIRANMENT AND SOCIAL MANAGEMENT
WOODEN PACKING SPECIFICATION SUBMISSION SHEET
(Attachment to the Tender Submission Sheet)
(To be filled by the Tenderer)

Form No. TPF 073-005

ISO 9001, ISO 14001 & ISO 45001 Certified

Contract No.

BREB SHIPPING BOX DESIGN NO. 1:

Applicable for all Line Hardwires ("B" Items), Conductor in Coils ("F" Items), Conductor Accessories ("E" Items), Underground Cable Accessories ("F" Items), Sectionalizing Devices (except Group Operated Air Breaker Switches and Reclosers) ("H" Items), Connectors ("I" Items), Meters and Metering Accessories ("J" Items), Street Light & Accessories ("L" Items), Miscellaneous ("ME" Items), Test Instruments ("TI" Items), Line Tools ("TL" Items), Capacitors and Accessories ("K" Items).

Technical Particulars	BREB Specification	Guaranteed Specification
1) Bidders' Name:	**To be mentioned.**	
2) Authorized Manufacturers' Name:	**To be mentioned.**	
3) Standard of manufacturing:	**Must be:** As per BREB Design of Respective Bid Document.	
4) BREB Items:	**Must be furnished:** The appropriate Design for the Items as per BREB Design of Respective Bid Document.	
5) Physical Requirements:	**Must be as follows:** The box sheathing shall be fabricated from Exterior Grade Plywood (**EGPW**) with a minimum thickness of three-eighth (3/8") or Sawn Board Lumber (**SBL**). All material and equipment shall be packed for export shipment to a tropical climate in wooden shipping boxes that are sufficiently durable to withstand numerous handing. The shipping boxes shall strong enough to prevent loss from pilferage or damage sustained from stacking, shipping or handling. Workmanship in the manufacture of the wooden shipping boxes shall be of the highest standard and the material used shall be in accordance with the best commercial practices. All lumber used shall be new, well seasoned, sound, and free of splits, decay or an excessive number of knots. Plywood used shall be new, free of splits and decay and shall be exterior grade quality. **Note:** "Sufficiently durable" means either the lumber shall be completely heartwood or mixture of sapwood and heartwood with preservative treatment of lumber with CCB having penetration of complete sapwood and treatable heartwood (if any) and retention of w/w 1.56% dry oxides of CCB in an assay zone of 0" to 0.62" as per requirement of REB. The sapwood of any timber species is not sufficiently durable. "Well seasoned" means lumber shall be kiln dried; otherwise the lumber shall not be well seasoned.	
6) Dimension of Box:	**Must be as follows:** The nominal dimensions of Shipping Box Design Number 1 shall be three-fourth (3/4) of a meter in height, three-fourth (3/4) of a meter in width and one meter in length (Height=0.75M, Width=0.75M, Length=1.0M). **Note that** these are nominal dimensions and that the Supplier may slightly vary these dimensions to better accommodate the packaged commodities.	
7) Thickness & Dimension of Sawn Lumbers (inch):	**Must be as follows:** **(1)** The minimum width of the sawn boards shall be six inches (6"). **(2)** The minimum thickness of the sawn boards utilized for the sides and roof of the box shall be five-eighth of an inch (5/8"), **(3)** Whereas the floorboards shall have a minimum thickness of one inch (1"). **(4)** The minimum dimensions of the skid runner and header shall be three inches (3") by four inches (4") and the skid runner shall be hardwood. **(5)** The minimum dimensions of the frames and cross-braces shall be one inch (1") by four inches (4").	
8) Box Preparation:	**Must be as follows:** (i) The box shall be strapped with four flat Steel/PVC strapping bands secured with crimped steel banding clips (ii) Two steel/PVC bands shall be installed vertically and two steel/PVC bands shall be installed horizontally (lengthwise). The minimum width of steel bands shall be three-fourth of an inch (3/4"). The width of PVC bands shall be 15mm (+/- 0.50 mm) and minimum thickness shall be 01 (one) mm. (iii) One-half inch (1/2") bolts shall be used to secure the header to the skid runner.	
9) Penalization formula for packing deviations detected during Post-Landing inspection in meet BREB SHIPPING BOX DESIGN NO.1	Must be as per BREB Board Decision no.13848 (Board Session-525) and BREB Instruction 500-30/PBS Instruction 100- 67	

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মোঃ রাকিব-উল-হাফিজ
সহঃ চিফ প্রোডাক্টস স্পেশালিষ্ট
পরিবেশ ও সামাজিক ব্যবস্থাপনা দপ্তর
বাপবিবোর্ড ঢাকা।

ISO 9001, ISO 14001 & ISO 45001 Certified

BANGLADESH RURAL ELECTRIFICATION BOARD, BANGLADESH
OFFICE OF THE ENVIRANMENT AND SOCIAL MANAGEMENT
WOODEN PACKING SPECIFICATION SUBMISSION SHEET
(Attachment to the Tender Submission Sheet)
(To be filled by the Tenderer)

Form No. TPF 073-006

Contract No---------

BREB SHIPPING BOX DESIGN NO. 2:

Applicable for all Transformers ("G" Items), Automatic Circuit Reclosers ("H" Items), Group Operated Air Break Switches ("H" Items), Automatic Voltage Regulators ("K" Items), Static Converters ("M" Items).

Technical Particulars	BREB Specification	Guaranteed Specification
1) Bidders' Name:	**To be mentioned.**	
2) Authorized Manufacturers' Name:	**To be mentioned.**	
3) Standard of manufacturing:	**Must be:** As per BREB Design of Respective Bid Document.	
4) REB Items:	**Must be furnished:** The appropriate Design for the Items as per BREB Design of Respective Bid Document.	
5) Physical Requirements:	**Must be as follows:** The box sheathing shall be fabricated from Exterior Grade Plywood **(EGPW)** with a minimum thickness of three-eighth (3/8") or Sawn Board Lumber **(SBL)**. All material and equipment shall be packed for export shipment to a tropical climate in wooden shipping boxes that are sufficiently durable to withstand numerous handing. The shipping boxes shall strong enough to prevent loss from pilferage or damage sustained from stacking, shipping or handling. Workmanship in the manufacture of the wooden shipping boxes shall be of the highest standard and the material used shall be in accordance with the best commercial practices. All lumber used shall be new, well seasoned, sound, and free of splits, decay or an excessive number of knots. Plywood used shall be new, free of splits and decay and shall be exterior grade quality. **Note:** "Sufficiently durable" means either the lumber shall be completely heartwood or mixture of sapwood and heartwood with preservative treatment of lumber with CCB having penetration of complete sapwood and treatable heartwood (if any) and retention of w/w 1.56% dry oxides of CCB in an assay zone of 0" to 0.62" as per requirement of REB. The sapwood of any timber species is not sufficiently durable. "Well seasoned" means lumber shall be kiln dried; otherwise the lumber shall not be well seasoned.	
6) Dimension of Box:	**Must be as follows:** There is no nominal dimension. The maximum number of pieces of equipment allowable per box will determine the size of the shipping boxes.	
7) Thickness & Dimension of Sawn Lumbers (inch):	**Must be as follows:** **(1)** Sides and Roof shall be 5/8" X 6" lumber (with 4" gap) or Exterior grade 3/8" plywood. **(2)** Frame shall 1" X 4" lumber. **(3)** Header shall 3" X 4" lumber. **(4)** Floor shall 1" X 6" lumber without gap. **(5)** Skid Runner shall 3" X 4" lumber.	
8) Box Preparation:	**Must be as follows:** (i) The box shall be strapped with three steel/PVC strapping bands and secured with crimped steel banding clips. (ii) Two bands installed vertically and one band installed horizontally (lengthwise); and the minimum width of steel bands shall be three-fourth of an inch (3/4"). The width of PVC bands shall be 15mm (+/-0.50 mm) and minimum thickness shall be 01 (one) mm. (iii) Six (6) One-half inch (1/2") bolts shall be used to secure the header to the skid runner. (iv) There shall be two (2) 1" X 4" notches in each skid.	
9) Penalization formula for packing deviations detected during Post-Landing inspection in meet BREB SHIPPING BOX DESIGN NO. 2	Must be as per BREB Board Decision no.13848 (Board Session-525) and BREB Instruction 500-30/PBS Instruction 100- 67	

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পরিবেশ ও সামাজিক ব্যবস্থপনা দপ্তর
বাপবিবোর্ড ঢাকা।

বাংলাদেশ পল্লী বিদ্যুতায়ন

ISO 9001, ISO 14001 & ISO 45001 Certified

BANGLADESH RURAL ELECTRIFICATION BOARD, BANGLADESH
OFFICE OF THE ENVIRANMENT AND SOCIAL MANAGEMENT
WOODEN PACKING SPECIFICATION SUBMISSION SHEET
(Attachment to the Tender Submission Sheet)
(To be filled by the Tenderer)

Form No. TPF 073-007

Contract No---------

BREB SHIPPING BOX DESIGN NO. 3:
Applicable for all Insulators ("C" Items)

Technical Particulars	BREB Specification	Guaranteed Specification
1) Bidders' Name:	**To be mentioned.**	
2) Authorized Manufacturers 'Name	**To be mentioned.**	
3) Standard of manufacturing:	**Must be:** As per BREB Design of Respective Bid Document.	
4) REB Items: C	**Must be furnished:** The appropriate Design for the Items as per BREB Shipping Box Design No. 3 (Revised January 2000)	
5) Physical Requirements:	**Must be as follows:** All material and equipment shall be packed for export shipment to a tropical climate in wooden shipping boxes that are sufficiently durable to withstand numerous handing. The shipping boxes shall strong enough to prevent loss from pilferage or damage sustained from stacking, shipping or handling. Workmanship in the manufacture of the wooden shipping boxes shall be of the highest standard and the material used shall be in accordance with the best commercial practices. All lumber used shall be new, well seasoned, sound, and free of splits, decay or an excessive number of knots. **Note:** "Sufficiently durable" means either the lumber shall be completely heartwood or mixture of sapwood and heartwood with preservative treatment of lumber with CCB having penetration of complete sapwood and treatable heartwood (if any) and retention of w/w 1.56% dry oxides of CCB in an assay zone of 0" to 0.62" as per requirement of REB. The sapwood of any timber species is not sufficiently durable. "Well seasoned" means lumber shall be kiln dried; otherwise the lumber shall not be well seasoned.	
6) Dimension of Box:	**Must be as follows:** The maximum number of pieces of equipment allowable per box will determine the size of the shipping boxes. For dimension of each C item, see Table-1 attached with Shipping Box Design No. 3	
7) Thickness of Battens and Separators (mm):	**Must be as follows:** For thickness of battens and separators in mm of each C item, see Table-1 attached with Shipping Box Design No. 3	
8) Box Preparation:	**Must be as follows:** i) All framing shall be located on the outside of the insulator box. (ii) Insulators shall be separated, both horizontally and vertically by wood partitions separators. (iii) Each insulator box shall be strapped with two flat Steel/PVC strapping bands, vertically installed. The minimum width of steel bands shall be one-half inch (1/2"). The width of PVC bands shall be 15mm (+/- 0.50 mm) and minimum thickness shall be 01 mm.	
9) Preservative Treatment:	**Must be as follows:** As per Foot Note No. 8 of Table-1 of Shipping Box Design No. 3 (Revised July 2000), all wooden battens and separators shall be treated with chemical preservatives according to the requirement of REB. to have preservative penetration of 100% sapwood and dry oxide retention of weight per weight (w/w) 1.56% in all treated zone equivalent to maximum thickness used in lumber.	
10) Penalization formula for packing deviations detected during Post- Landing inspection in meet BREB SHIPPING BOX DESIGN NO.3	Must be as per BREB Board Decision no.13848 (Board Session-525) and BREB Instruction 500-30/PBS Instruction 100- 67	

06.08-2020
(মোঃ রকিব-উল-হাফিজ)
[illegible] প্রোডাক্টস স্পেশালিষ্ট (চলতি)
বাপবিবো, ঢাকা

বাংলাদেশ পল্লী বিদ্যুতায়ন

ISO 9001, ISO 14001 & ISO 45001 Certified

Form No. TPF 073-004

BANGLADESH RURAL ELECTRIFICATION BOARD, BANGLADESH
OFFICE OF THE ENVIRANMENT AND SOCIAL MANAGEMENT
SPECIFICATION SUBMISSION SHEET
(Attachment to the Tender Submission Sheet)
(To be filled by the Tenderer)

Contract No> ------------

<u>WOODEN CONDUCTOR REELS</u> (Applicable for all "D" and "N" Items)

THE BIDDER'S CONDUCTOR MANUFACTURER MUST FURNISH THE FOLLOWING INFORMATION AS SUPPLIED BY THE WOOD PRODUCTS TREATMENT PLANT WHO WILL SUPPLY REELS AND LAGGINGS FOR THIS SUPPLY CONTRACT:

1.NAME, LOCATION AND ADDRESS OF THE TREATMENT PLANT:..	7.NAME OF THE OFFERED TIMBER SPECIES FROM THE REB LIST ONLY TO BE USED FOR THE CONSTRUCTION OF REELS /BOXES (BOTH COMMON AND SCIENTIFIC NAMES):.. ....
2.PRESSURE TO BE USED IN TREATING THE REELS/BOXES FOR THIS CONTRACT:................................(Kg/cm²)	
3.DRYING METHOD TO BE USED IN DRYING THE LUMBER TO BE USED IN REELS, LAGGINGS /BOXES:........................	8.NAME OF THE OFFERED PRESERVATIVE FROM THE REB LIST ONLY TO BE USED FOR TREATING LUMBERS (TRADE NAMES ARE NOT ELIGIBLE): ....
4.MOISTURE CONTENT TO BE ATTAINED WITH FURNISHED DRYING METHOD:..	9. OFFERED PRESERVATIVE PENETRATION: ....(mm)
5.PROVIDE INFORMATION REGARDING THE TESTING FACILITIES AVAILABLE IN THE TREATMENT PLANT THAT WILL BE USED TO DETERMINE PRESERVATIVE PENETRATION, RETENTION AND MOISTURE CONTENT BEFORE TREATMENT:..	10.OFFERED PRESERVATIVE RETENTION: ....(W/W%)
6.DESCRIBE OTHER PRESSURE TREATED WOOD PRODUCTS PRODUCED AND SOLD BY THIS PLANT:........................	11. OFFERED DESIGN NO. OF REELS/BOXES FROM THE REB LIST ONLY:.. ....

12. Wooden reel preservative treatment requirement: Meet "American Wood Preserve Associated Standards" and "BREB Requirements".

13.Penalization Formula for Deviations Detected During Post- Landing inspection in WOODEN CONDUCTOR REELS, (All "D" and "N" Items). Must be as per BREB Board Decision no.13848 (Board Session 525) and BREB Instruction 500-30/PBS Instruction 100- 67. **Meet "American Wood Preserve Associated Standards" and "BREB Requirements".**

WE CERTIFY THAT WE ARE AWARE OF THE BREB SPECIFICATION UNDER THE IDENTIFIED AS ....BID PACKAGE ....SUB-PACKAGE.............................. THE INFORMATION PROVIDED ABOVE IS CORRECT TO THE BEST OF OUR KNOWLEDGE AND IS SUBJECT TO VERIFICATION. WOOD REELS AND THEIR LAGGINGS/BOXES SHALL BE CONSTRUCTED FROM NEW LUMBER WHICH SHALL BE SQUARE SAWN, BE OF SMOOTH SURFACE, WITH NO SPLITS, WARPS, CROOKS, LOOSE FIBERS, DECAY OR INSECT INFESTATION. THE LUMBER USED SHALL BE PRESERVATIVE TREATED BY PRESSURE PROCESS TO ATTAIN PENETRATION AND RETENTION REQUIREMENTS AS STIPULATED IN REB'S STANDARD PUBLICATIONS..AND THE LATEST REVISION THEREOF.

NAME, SIGNATURE, TITLE, DATE AND SEAL OF THE BIDDER:	NAME, SIGNATURE, TITLE, DATE AND SEAL OF THE MNUFACTURER:	NAME, SIGNATURE, TITLE, DATE AND SEAL OF THE WOOD PRESERVING PLANT:
NAME PRINTED IN:....................................	NAME PRINTED IN:....................................	NAME PRINTED IN:....................................
AUTHORIZED SIGNATURE:....................................	AUTHORIZED SIGNATURE:....................................	AUTHORIZED SIGNATURE:....................................
TITLE AND DATE PRINTED IN:....................................	TITLE AND DATE PRINTED IN:....................................	TITLE AND DATE PRINTED IN:....................................
SEAL:	SEAL:	SEAL:

THE BIDDER'S MANUFACTURING PLANT OR THE PLANT THAT WILL SUPPLY CONDUCTOR/INSULATOR/TRANSFORMER AND HIS SUB-CONTRACTOR THAT WILL FABRICATE THE REELS, LAGGINGS ON/IN WHICH THE GOODS WILL BE ARRANGED MUST ENCLOSE WITH THIS PROPOSAL A PRINTED ORIGINAL COPY OF BOOKLETS, BROCHURES, ANY AND ALL OTHER ORIGINAL INFORMATION THAT WILL PERMIT A TRUE AND UNBIASED OPINION OF ITS CAPABILITIES TO PRODUCE REELS, LAGGINGS IN CONFORMANCE WITH REB'S STANDARDS AND SPECIFICATIONS.

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সহঃ টিম্বার প্রোডাক্টস স্পেশালিষ্ট
পরিবেশ ও সামাজিক ব্যবস্থাপনা দপ্তর
বাপবিবোর্ড ঢাকা